

Markov Blanket Optimization with Graph Diffusion

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Preliminary

The Markov blanket reconstruction serves the only goal: it reduces the computational complexity of the causal graph reconstruction. The model selection complexity is $C = 2^n - 1$, where n is the number of features or model parameters. The structural causal model selection complexity is the number of the selected edges given the number of nodes. For n nodes the number of the edges is $m = 2(n^2 - n)$.

The goal of the proposed algorithm is a DAG of minimal complexity. It reduces the SCM causal graph reconstruction problem to the series of problems of lesser complexity. Instead of the graph reconstruction one has to reconstruct the Markov blanket. The nodes and edges around the target variable of the Markov blanket form a smaller causal graph. The whole causal graph is a union of the Markov blankets.

The main problem of model selection remains in SCM selection. It is the problem of high correlation between parents of a variable. It is difficult to discover causality when one observes high dependency between the parents. Note that when the SCM selection criterion is accuracy, averaging features with similar distribution boost the accuracy.

There given

The graph $G = (V, E)$, the probabilities of vertices $P(V)$, the probabilities of edges $P(E)$, the data D , and the class of SCM with the Bayesian evidence $p(D|\mathbf{A}; \mathbf{w})$

The initial set of blankets $\{B\}$, where each initial blanket is an isolated vertex, $B = X$: it has no parents, spouses and children.

Algorithm of the graph reconstruction

1. Random sampling. Select a vertex V as a target Y from the set of vertices in G .
2. Estimate the blanket B with an algorithm like PCMB.
3. Random sampling. Append a random item $P(V)$ to the blanket B and test it.
4. Insert the blanket into the graph $G = \bigcup B$.
5. Fine-tune the SCM model the local parameters $\mathbf{b}$ and hyper-parameters $\mathbf{A}$.
6. Compute the model evidence $p(D|\mathbf{A}; \mathbf{w})$. Stop when it converges.